Who Owns Whom?

Predicting batter-pitcher matchups from career lines, head-to-head history and archetypes

Trent Reimers | MLB 2020-2026 | October 16, 2026

The question. When a batter faces a pitcher, which history predicts the outcome best: each player's **career** line, their **head-to-head** (H2H) history, or how their **archetypes** match up?

The answer. Career lines, and nothing improves on them. Predicting every 2025–2026 PA from 2020–2024 history, 45 comparisons of archetype, H2H and talent-tier adjustments against career alone find 0 reliably better and 0 reliably worse. A pair's own head-to-head history is far worse than the league average on its own (15% more error for wOBA) and adds nothing on top of career.

What repeats, but doesn't forecast. Archetype matchups are real patterns, not pure noise: 31 of 120 cells clear $|z| > 2$ in 2020–2024, and about 33% of each cell's wOBA effect (51% for K%) shows up again in 2025–2026. A few targeted pitch-level pairings repeat strongly. These effects are too small, relative to plate-appearance luck, to sharpen a prediction. *The matched-pair results are exploratory.*

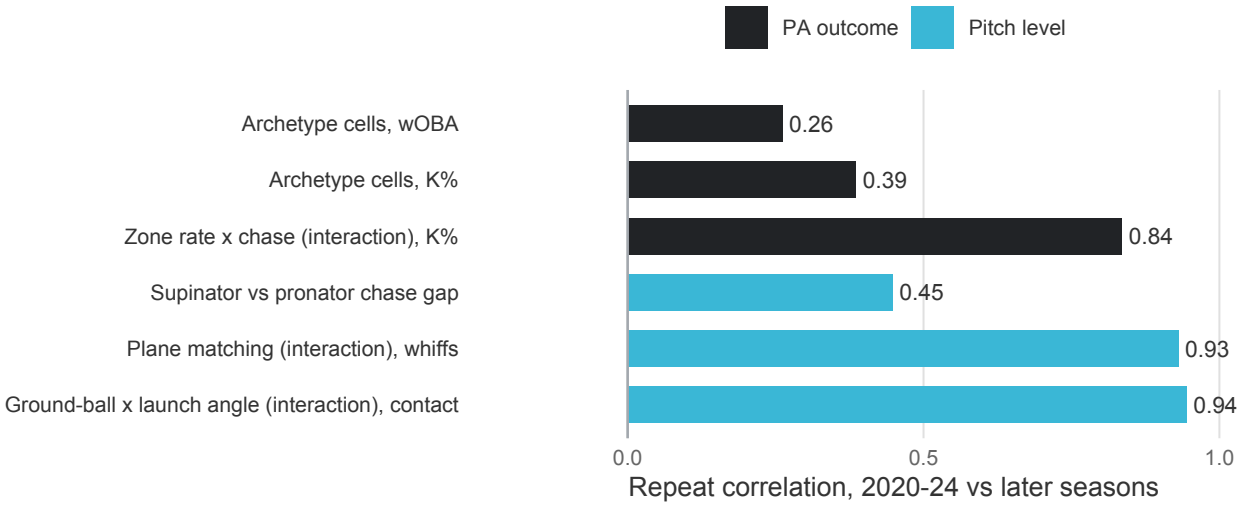


Figure 1: At a glance: how strongly each effect measured in 2020-24 shows up again in the next two seasons (weighted correlation across cells). Repeating is not the same as improving a forecast: none of these beats career lines in Part 3.

PART 1

Question and design

Three kinds of history

When Rafael Devers steps in against Gerrit Cole, there are three things we could look at:

1. **Career stats**: Devers' career line against righties and Cole's career line against lefties.
2. **Head-to-head**: what Devers has done against Cole specifically.
3. **Archetype matchup**: how *batters like Devers* have done against *pitchers like Cole*.

Broadcasts lean on number 2 and front offices increasingly talk about number 3. This report tests all three and hybrids of them out of sample.

Data and designs

Data. MLB Statcast regular season 2020–2026: 1.16M PAs after dropping intentional walks, sacrifice bunts, catcher's interference and truncated PAs. Outcomes: wOBA, xwOBA (expected wOBA from exit velocity and launch angle, actual values for K and BB), K%, BB%, and a six-way log loss over K / BB+HBP / 1B / 2B+3B / HR / out (how well predicted outcome probabilities match what happened; lower is better).

Table 1: Three test designs.

Design	Test set	What it asks
B. Next two seasons (main test)	Every 2025-2026 PA (364,263), predicted from 2020-2024 only	Does anything beat career lines, including for thin careers?
A. High-volume pairs	352 batter-pitcher pairs with 30+ meetings; second half of each (6,278 PAs)	Best case for H2H: every pair has real history
C. Matched pairs (exploratory)	Same as B, for five pitcher-trait x batter-trait grids	Do targeted pairings show interactions that last?

No peeking. Test PAs never feed any history. In design B, everything comes from 2020–2024. In design A, pairs are grouped by the season their test half starts, and each group's career lines, archetypes and cells come only from earlier seasons; a pair's own first-half meetings count only as its H2H. All regression weights, including how hard career lines are regressed and how stacked adjustments combine, are tuned on a separate validation season (2024 from 2020–2023) with design A's test PAs removed. Confidence intervals resample batters *and* pitchers, since PAs share both.

Table 2: Main methods. wOBA and xwOBA combine additively; outcome rates combine with log5. Five more variants are in the appendix.

Method	How it predicts
League (platoon)	League rate for same-hand or opposite-hand matchups (the baseline)
Career (individuals)	Batter's career vs that hand + pitcher's career vs that side, regressed to league
Career shrunk to velocity / bat-speed tier	Same, but regressed toward the player's velocity tier (pitchers) or bat-speed tier (batters)
H2H raw	The pair's own history, as is
Career + H2H	Career + the pair's gap from career, regressed
Archetype matchup	Rate in the batter-archetype x pitcher-archetype x batter-side cell
Career + archetype	Career + the cell's gap from what its players' careers predict
Career + batter vs similar pitchers	Career + this batter's gap vs the 25 most similar pitchers
All archetype info	Archetype prior + cell + batter- and pitcher-specific archetype gaps, weights tuned jointly

PART 2

The archetypes

Pitchers

Built from 2020–2024 pitches, separately for RHP and LHP, from primary-fastball velocity, median arm angle and **four-seam spin efficiency** (200+ pitches). Rules, in order: **Soft Tosser** (fastball under 91.5 mph); **Stock Arm** (arm angle 33–47°); otherwise **High-Slot** or **Low-Slot**, then **Pronator** (four-seam active spin $\geq$ 90%) or **Supinator** (below, or no real four-seamer).

Why spin efficiency. Spin efficiency is the share of spin that moves the ball rather than gyro spin that doesn't. Pronators get behind the ball and throw true-ride four-seamers at 90–100% efficiency, usually with a good changeup. Supinators release with an open wrist that tilts the axis toward gyro: cut-ride four-seamers, sweepers, gyro sliders, seam-shifted sinkers. The number is Baseball Savant's **Active Spin** (one value per pitcher-season, pooled by four-seam count); cutter- or sinker-only pitchers count as supinators. It correlates only weakly with arm angle ($r = 0.24$).

Table 3: Pitcher archetypes from 2020-2024 (medians). No 4S = too few four-seamers to measure.

Archetype	N	FB velo	Slot	4S eff.	No 4S	Examples (most pitches)
LHP High-Slot Pronator	35	93.0	50	97%	0	Blake Snell, Carlos Rodón
LHP High-Slot Supinator	17	93.3	51	87%	2	Max Fried, Drew Smyly
LHP Low-Slot Pronator	20	93.5	28	94%	0	Sean Manaea, Andrew Heaney
LHP Low-Slot Supinator	27	93.5	26	83%	15	Trevor Rogers, Alex Wood
LHP Soft Tosser	107	89.8	40	91%	28	Tyler Anderson, Kyle Freeland
LHP Stock Arm	105	93.5	41	92%	14	Patrick Corbin, Framber Valdez
RHP High-Slot Pronator	111	94.4	52	96%	0	Logan Gilbert, Nick Pivetta
RHP High-Slot Supinator	73	94.5	53	86%	18	Dylan Cease, Sonny Gray
RHP Low-Slot Pronator	64	95.3	27	95%	0	Aaron Nola, Max Scherzer
RHP Low-Slot Supinator	126	94.3	28	82%	45	Luis Castillo, Logan Webb
RHP Soft Tosser	123	90.1	37	88%	35	Marcus Stroman, Kyle Hendricks
RHP Stock Arm	408	94.4	40	92%	46	Corbin Burnes, Zack Wheeler

Batters

A batter needs 75+ PAs (z-scores scaled against hitters with 250+). Each gets four scores and takes the highest if it clears 0.75; everyone else is **Balanced**. **Pull Power**: EV90, pulled fly balls and liners, ISO. **Slap Hitter**: low K%, whiff%, ISO and EV90. **Walk Machine**: high BB%, low chase. **Free Swinger**: high chase and whiff. Pitchers and batters are also split into velocity and bat-speed thirds for the talent-tier prior.

Table 4: Batter archetypes from 2020-2024 (medians).

Archetype	N	K%	BB%	ISO	Chase%	EV90	Examples (strongest fit)
Balanced	339	24%	8%	0.141	29%	103.6	Didi Gregorius, Jorge Mateo
Free Swinger	138	31%	7%	0.151	35%	104.8	Trey Cabbage, Jack López
Pull Power	84	26%	10%	0.221	29%	106.6	Aaron Judge, Byron Buxton
Slap Hitter	134	16%	8%	0.099	29%	100.5	David Fletcher, Luis Arraez
Walk Machine	167	27%	12%	0.156	22%	103.9	Alex Avila, Juan Soto

PART 3

Predicting the next two seasons

Nothing beats career

Table 5: Design B: skill vs league on every 2025-2026 PA (drop in error). Single PAs are mostly luck, so all skills are small; the comparison with career is what matters.

Method	wOBA	xwOBA	K%	BB%	Log loss
League (platoon)	0.00%	0.00%	0.00%	0.00%	0.00%
Career (individuals)	**0.13%**	**0.46%**	**1.50%**	**0.51%**	**0.89%**
Career shrunk to velocity / bat-speed tier	0.11%	0.44%	1.52%	0.49%	0.89%
H2H raw	-14.94%	-14.59%	-15.02%	-14.71%	-68.72%
Career + H2H	0.13%	0.46%	1.49%	0.51%	0.89%
Archetype matchup	0.02%	0.07%	0.40%	0.14%	0.28%
Career + archetype	0.14%	0.46%	1.49%	0.51%	0.89%
Career + batter vs similar pitchers	0.13%	0.46%	1.50%	0.51%	0.89%
All archetype info	0.12%	0.43%	1.51%	0.49%	0.89%

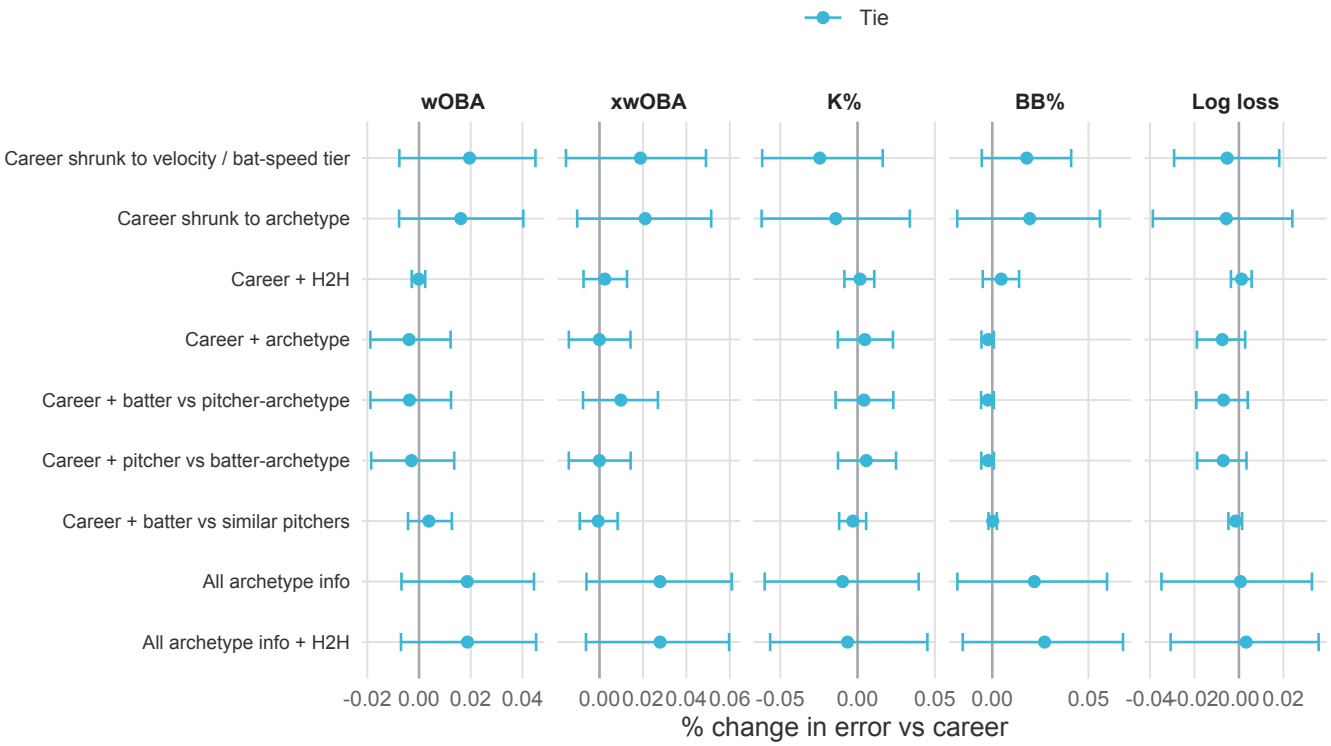


Figure 2: Design B: change in error vs career as a share of the league baseline's error, 95% intervals resampling batters and pitchers. Right of zero = worse than career alone.

With 364,263 PAs, this design can see small effects. Every adjustment ties career: 0 of 45 comparisons are reliably better and 0 reliably worse. That includes the stacked "all archetype info" model once its pieces are tuned together. (Summing separately tuned pieces made it look worse in an earlier draft; that was double counting, not a real cost of archetypes.)

Archetype effects partly last

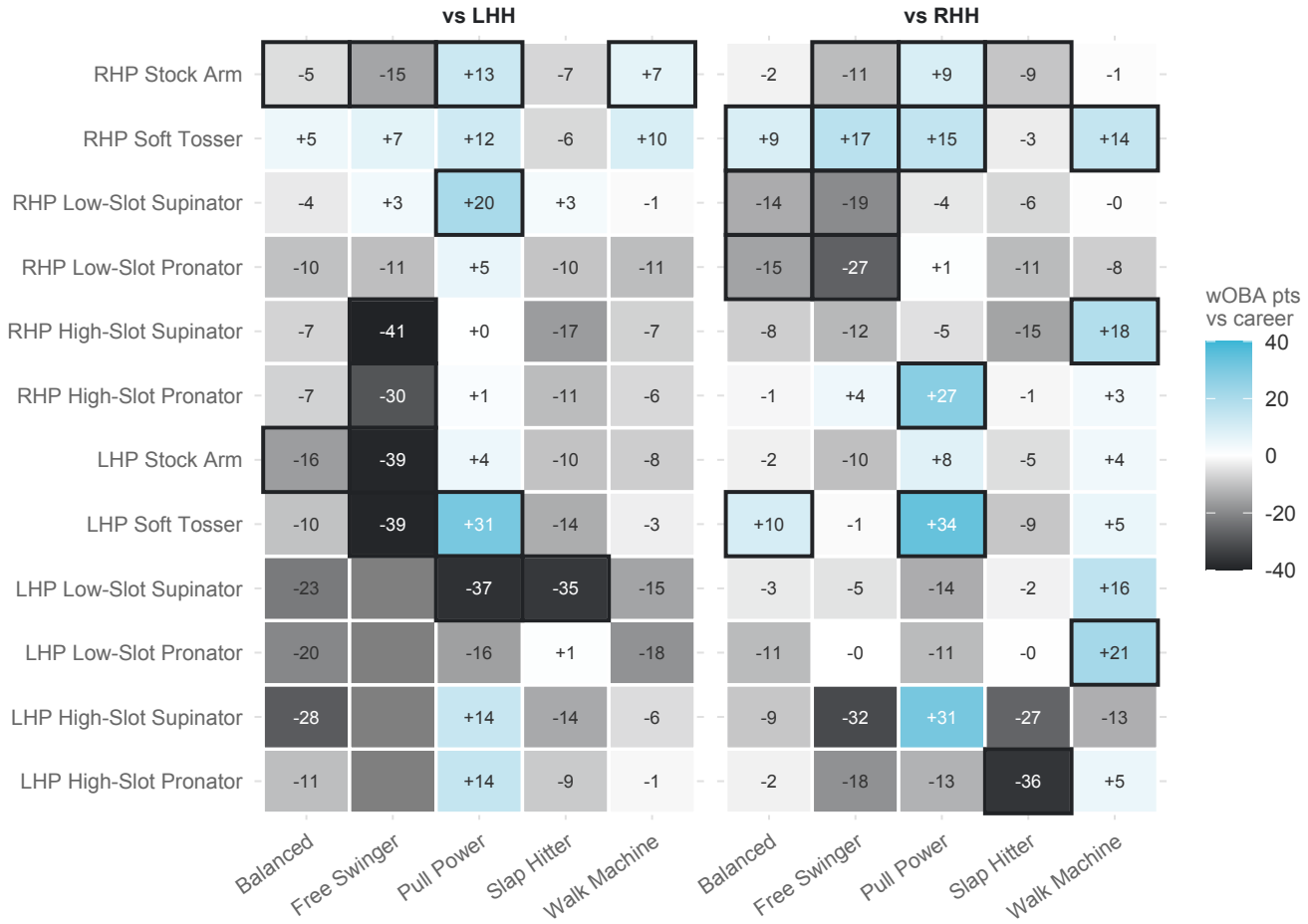


Figure 3: wOBA in each archetype cell minus what its players' careers predict, 2020-2024. Cyan = batters outperform, charcoal = pitchers. Outlined: $|z| > 2$. Blank: under 500 PAs.

Table 6: Each cell's 2020-2024 effect vs the same cell in 2025-2026.

Stat	Cells	Correlation	Share that carries forward
wOBA	120	0.26	33% (SE 11%)
xwOBA	120	0.13	18% (SE 12%)
K%	120	0.39	51% (SE 11%)
BB%	120	0.25	30% (SE 11%)

Real, but small. 31 of 120 cells clear $|z| > 2$ in 2020–2024 (about 5 expected by chance; allowing for repeated players barely changes the standard errors). About 33% of a cell's wOBA effect and 51% of its K% effect carry into the next two seasons. But a 20-point wOBA cell effect that only partly carries forward is worth a few points in practice, under a tenth of a hit across the 10–15 PAs two players see in a season, which is why adding it to career lines doesn't measurably help.

Thin careers

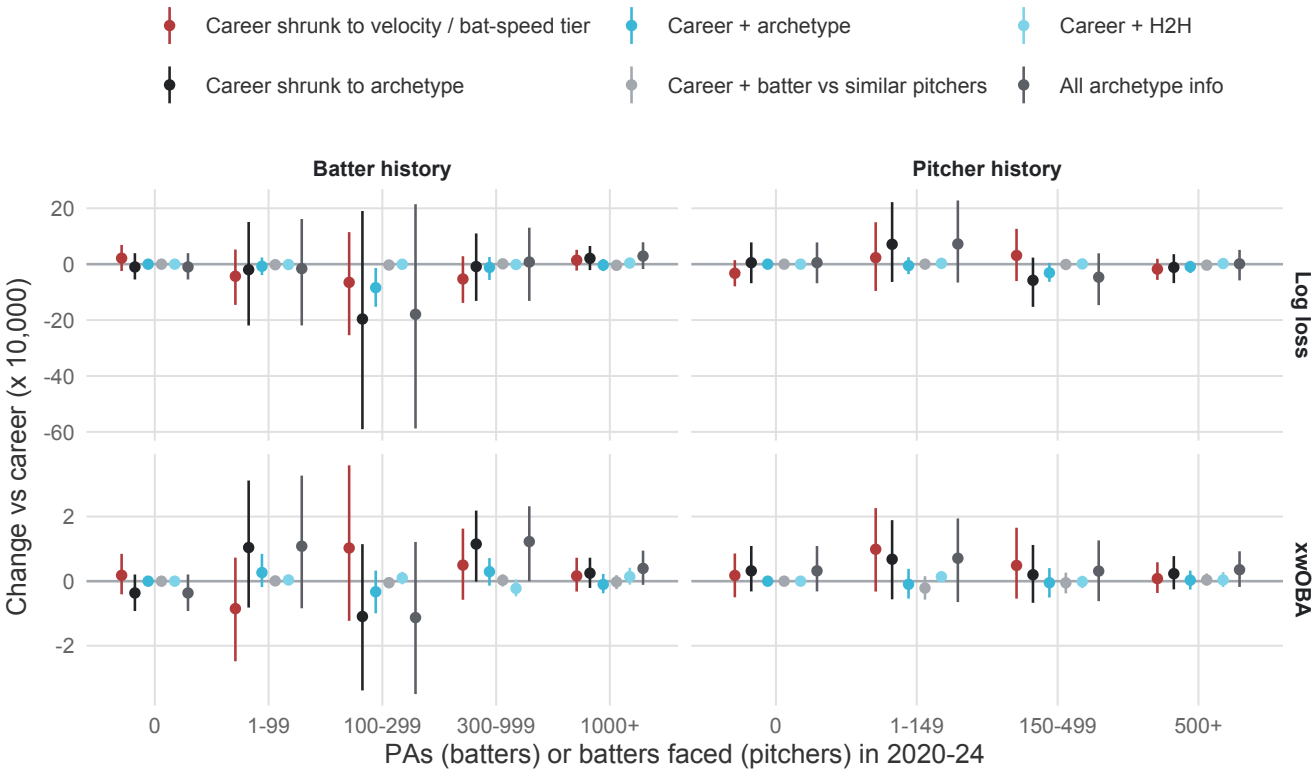


Figure 4: Design B: change in per-PA error vs career by 2020-24 history, 95% intervals resampling batters and pitchers. Below zero = better than career.

Splitting the test PAs by how much history each player had, 2 of 135 comparisons beat career and 0 are worse, about what chance produces. Archetypes and talent tiers don't rescue thin careers here. A caveat: players with almost no history can't be classified either, so this test is weak at the thin end; a version that updates game by game would test it properly.

Which archetype definition lasts?

Table 7: Archetype variants rebuilt from 2020-2024: correlation of each cell's history effect with its effect in 2025-2026.

			Cell effects carry forward		
Variant	**Cells**	**|z|>2**	**wOBA**	**xwOBA**	**K%**
Main (spin cut 90%, pooled labels)	120	31	0.26	0.13	0.39
No pronator/supinator split (slot only)	80	30	0.31	0.16	0.45
Spin cut 85%	120	33	0.28	0.14	0.36
Spin cut 95%	120	35	0.27	0.16	0.34
Spin-efficiency thirds	160	38	0.22	0.11	0.28
Per-season pitcher labels	120	31	0.31	0.18	0.40
Adds Long Uphill Swing batters	144	32	0.22	0.14	0.35

The most durable definition for wOBA is "Per-season pitcher labels". Re-labeling pitchers each season helps (they change slots and add pitches), and dropping the pronator/supinator split loses nothing at the PA level. No variant makes career + archetype beat career.

PART 4

Velocity and bat speed as a prior

Tested directly, it ties

Table 8: Design B: career regressed toward the player's velocity or bat-speed third, minus career regressed toward league (per-PA error; negative = better).

Stat	Change vs career	95% interval	Verdict
wOBA	+5.14e-05	[-2.00e-05, +1.18e-04]	Tie
xwOBA	+2.56e-05	[-2.10e-05, +6.66e-05]	Tie
K%	-4.24e-05	[-1.07e-04, +2.82e-05]	Tie
BB%	+1.54e-05	[-4.71e-06, +3.53e-05]	Tie
Log loss	-7.64e-05	[-4.20e-04, +2.62e-04]	Tie

An earlier draft recommended regressing thin careers toward a velocity- or bat-speed-aware average, because power arms and fast bats beat their league-regressed career lines in the matched-pair grid. Those tier averages do repeat (velocity x bat speed cells repeat at $r = 0.84$ for xwOBA, almost all of it the tier averages rather than the pairing). But tested directly as a prior, it **ties** career on every stat overall, and by history bucket only 1 of 27 comparisons are reliably better (0 worse). Once career lines are regressed at the tuned strength, knowing a player's velocity or bat speed adds nothing measurable to the forecast. The recommendation is withdrawn.

PART 5

High-volume matchups

The best case for head-to-head

Table 9: Design A: skill vs league, second half of 352 pairs with 30+ meetings (history from before each split only).

Method	wOBA	xwOBA	K%	BB%
League (platoon)	0.0%	0.0%	0.0%	0.0%
Career (individuals)	**5.4%**	**13.0%**	**19.9%**	**11.0%**
Career shrunk to velocity / bat-speed tier	5.6%	12.9%	19.8%	11.0%
H2H raw	-87.0%	-50.9%	-30.0%	-52.9%
Career + H2H	5.4%	12.9%	21.0%	12.2%
Archetype matchup	0.4%	5.0%	-1.0%	4.8%
Career + archetype	5.6%	13.9%	19.6%	11.0%
Career + batter vs similar pitchers	5.4%	13.0%	20.2%	10.8%
All archetype info	3.2%	12.7%	19.2%	10.6%

Even with up to 30 earlier meetings, a pair's own record (87% worse than league for wOBA) loses badly, and adding it to career ties career. With about 17 test PAs per pair, luck dominates: chance alone accounts for an expected error of roughly 0.127 for wOBA (an approximate estimate), nearly all of the league baseline's 0.128. Career cuts that to 0.125, which leaves little room for anything else. Of 36 adjustments compared with career, 0 are reliably better and 0 reliably worse.



Figure 5: Design A: error vs how many PAs of prior are blended with the pair's H2H. Left = trust H2H fully, right = ignore it.

Cross-validation over pairs picks how much H2H to add on top of career. For wOBA it mostly picks "ignore" (1e+05 PAs of prior: 0% weight on 17 meetings); for K% it picks 150 (10% weight).

Table 10: Design A: correlation [95% CI] between each source's raw gap from career and the second-half gap from career.

Source	wOBA	xwOBA	K%	BB%
Archetype interaction	0.01 [-0.18, 0.20]	0.05 [-0.10, 0.22]	-0.03 [-0.20, 0.16]	0.07 [-0.11, 0.26]
Batter vs pitcher-archetype	0.11 [-0.06, 0.28]	0.05 [-0.12, 0.23]	0.08 [-0.07, 0.25]	-0.00 [-0.19, 0.21]
Batter vs similar pitchers	0.06 [-0.11, 0.23]	0.01 [-0.19, 0.19]	0.07 [-0.09, 0.23]	-0.13 [-0.29, 0.03]
H2H first half	0.00 [-0.20, 0.18]	0.06 [-0.11, 0.22]	0.10 [-0.07, 0.28]	0.12 [-0.07, 0.33]
Velocity / bat-speed prior	0.05 [-0.14, 0.22]	0.03 [-0.12, 0.18]	0.05 [-0.14, 0.24]	0.04 [-0.15, 0.22]

0 of 20 intervals clear zero: no source knows anything about these matchups that career lines don't.

Table 11: The 8 hottest and 8 coldest first-half H2H lines vs what happened next.

Batter	Pitcher	H2H PA	H2H wOBA	Career pred	Test PA	Test wOBA
Batter 'owned' the pitcher						
Paul Goldschmidt	Marcus Stroman	15	0.743	0.345	15	0.467
Bobby Witt Jr.	Bailey Ober	15	0.707	0.321	16	0.459
Rafael Devers	Gerrit Cole	20	0.715	0.358	20	0.547
Ian Happ	Mitch Keller	16	0.672	0.360	17	0.441
Jose Altuve	Nathan Eovaldi	15	0.677	0.375	15	0.373
Bo Bichette	Nick Pivetta	17	0.635	0.338	17	0.188
Gunnar Henderson	Zack Littell	16	0.641	0.353	16	0.078
Adley Rutschman	José Berríos	16	0.650	0.364	16	0.656
Pitcher 'owned' the batter						
Matt Olson	Sandy Alcantara	15	0.060	0.328	16	0.188
Trevor Story	Kevin Gausman	15	0.060	0.317	16	0.222
Bobby Witt Jr.	Joe Ryan	18	0.089	0.329	19	0.547
TJ Friedl Jr.	Mitch Keller	16	0.100	0.339	17	0.241
J.D. Martinez	Jordan Montgomery	16	0.100	0.336	17	0.294
Max Muncy	Yu Darvish	15	0.107	0.334	15	0.397
Josh Rojas	Yu Darvish	15	0.107	0.319	15	0.343
Marcus Semien	Logan Gilbert	20	0.135	0.345	20	0.225

The "owned" batters went from 0.680 to 0.404 (career prediction 0.352), keeping about 16% of their gap, within noise for about 130 PAs. The "owned" pitchers' batters went from 0.095 to 0.309 (career 0.331).

Table 12: Design A skill vs league by PA floor, each floor rebuilt from scratch (pairs: 20+ = 2024, 25+ = 803, 30+ = 352, 35+ = 151, 40+ = 66).

	wOBA					xwOBA				
Method	**20+**	**25+**	**30+**	**35+**	**40+**	**20+**	**25+**	**30+**	**35+**	**40+**
League (platoon)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Career (individuals)	3.1%	3.6%	5.4%	6.1%	2.5%	10.1%	10.7%	13.0%	14.8%	10.4%
Career shrunk to velocity / bat-speed tier	2.9%	3.4%	5.6%	5.6%	2.4%	9.9%	10.4%	12.9%	13.8%	8.7%
H2H raw	-100.2%	-86.7%	-87.0%	-93.1%	-94.8%	-73.2%	-65.7%	-50.9%	-49.2%	-46.9%
Career + H2H	3.1%	3.6%	5.4%	6.1%	2.0%	10.2%	10.6%	12.9%	14.5%	7.0%
Archetype matchup	1.1%	1.2%	0.4%	3.3%	4.8%	3.9%	3.4%	5.0%	9.8%	11.0%
Career + archetype	3.6%	3.9%	5.6%	6.8%	3.5%	10.7%	11.2%	13.9%	16.2%	11.8%
Career + batter vs similar pitchers	3.1%	3.5%	5.4%	5.8%	1.4%	10.2%	10.4%	13.0%	15.1%	9.8%
All archetype info	3.0%	3.3%	3.2%	5.7%	1.8%	10.4%	10.8%	12.7%	15.9%	11.1%

PART 6

Matched pairs (exploratory)

Pair the pitcher trait with the batter trait it attacks

The archetype grid mixes many traits into each label. A sharper probe pairs **one pitcher trait with the batter trait it should attack**, splits each into thirds using 2020–2024 only, and asks whether the 3 × 3 pattern repeats in 2025–2026. Each cell is split into *tier averages* (does a power arm beat his career line against everyone?) and the *pure interaction* (does it depend on whom he faces?); only the interaction is a matchup effect.

Read this part as hypothesis generating. Each interaction rests on about 4 independent numbers, the outcomes within a pair overlap (PA xwOBA, contact xwOBA and wOBA share the same contact), and the three pairs highlighted below were picked after seeing all 35 results. Under no effect, 1–3 correlations above 0.85 are expected by chance; 5 appear. The highlighted patterns are worth re-testing on 2027 data, not acting on.

Table 13: Five matched pairs; tiers are thirds of qualified 2020-24 players.

Pair	Pitcher	Batter
Approach angle x swing plane	Four-seam approach angle vs league at that height	Attack angle on contact (bat tracking)
Velocity x bat speed	Primary fastball velocity	Bat speed (bat tracking)
Zone rate x chase	Share of pitches in the zone	Chase rate
Ground-ball pitcher x launch angle	Ground balls / balls in play	Average launch angle
Fastball usage x fastball hitter	Share of fastballs	Fastball vs spin whiffs and contact quality



Figure 6: How well each pair's pure interaction repeats from 2020-24 to later seasons (weighted correlation across the 9 cells).

Like-on-like batted balls favor the pitcher

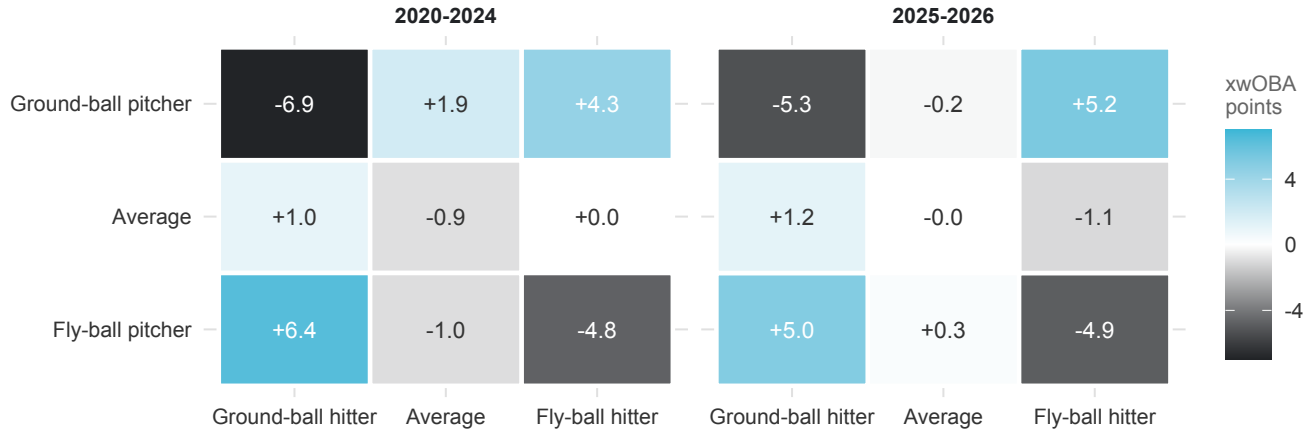


Figure 7: Ground-ball pitcher x launch angle: interaction in contact xwOBA (points). Cyan = batter gains, charcoal = pitcher gains.

Ground-ball pitcher vs ground-ball hitter, or fly-ball vs fly-ball, takes about 5–7 points of contact xwOBA off the hitter; opposite pairings give it back. It repeats at $r = 0.94$ at the pitch level and 0.99 for PA xwOBA, the strongest and most physically sensible pattern here.

Whiffs drop when the swing plane matches the pitch plane

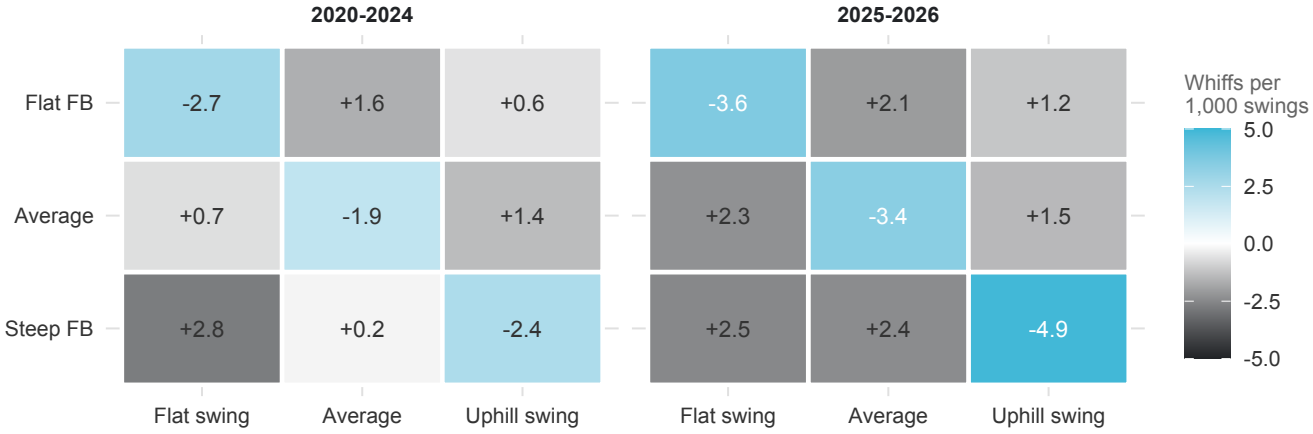


Figure 8: Fastball approach angle x swing plane: interaction in four-seam whiff rate (per 1,000 swings). Charcoal = more whiffs.

Flat four-seamers get fewer whiffs from flat swings and steep ones fewer from uphill swings. It repeats ($r = 0.93$) but is about half a point of whiff rate and doesn't reach K% (interaction r = -0.65).

Nibblers out-strike chasers

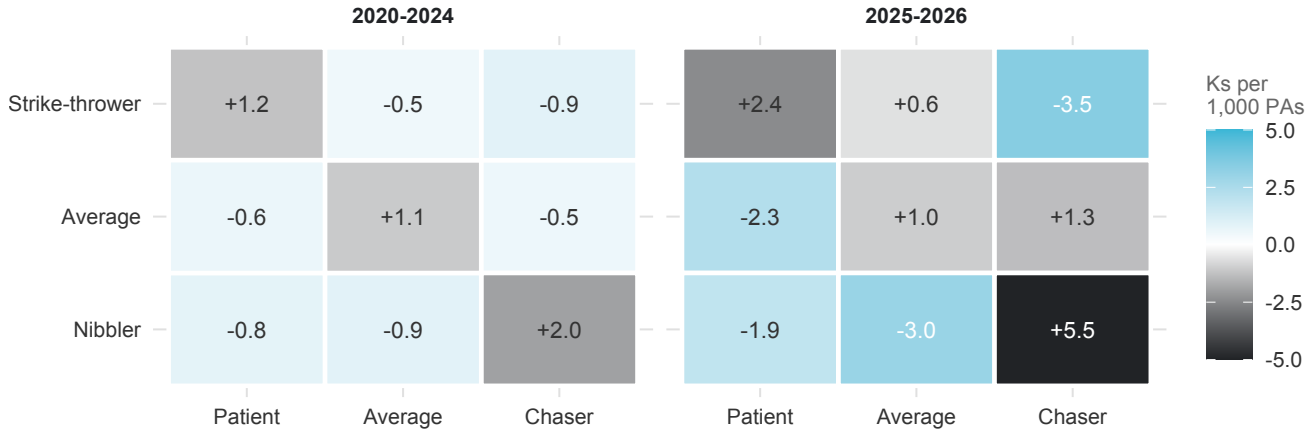


Figure 9: Zone rate x chase rate: interaction in K% (per 1,000 PAs). Charcoal = more strikeouts.

Nibblers get extra strikeouts from chasers and strike-throwers from patient hitters ($r = 0.84$ for K%, 0.76 for pitch-level whiffs), with a larger pattern in the later seasons than in history.

Table 14: Change in test-season error vs career (% of career's error, 95% interval resampling batters and pitchers) when each pair's history cell effect is added at full weight. Bold = reliably better; red = reliably worse.

Pair	Stat	Full cell	Interaction only
Approach angle x swing plane	K%	+0.004 [-0.026, +0.037]	+0.002 [-0.002, +0.006]
Approach angle x swing plane	xwOBA	-0.001 [-0.017, +0.015]	-0.000 [-0.004, +0.004]
Fastball usage x fastball hitter	K%	+0.008 [-0.012, +0.027]	+0.003 [-0.009, +0.014]
Fastball usage x fastball hitter	xwOBA	-0.003 [-0.017, +0.011]	+0.001 [-0.011, +0.012]
Ground-ball pitcher x launch angle	K%	+0.017 [-0.003, +0.036]	+0.000 [-0.005, +0.005]
Ground-ball pitcher x launch angle	xwOBA	-0.002 [-0.021, +0.017]	-0.004 [-0.016, +0.009]
Velocity x bat speed	K%	-0.021 [-0.058, +0.014]	+0.001 [-0.001, +0.003]
Velocity x bat speed	xwOBA	-0.028 [-0.055, +0.000]	+0.000 [-0.006, +0.006]
Zone rate x chase	K%	**+0.010 [+0.001, +0.018]**	-0.002 [-0.006, +0.001]
Zone rate x chase	xwOBA	-0.004 [-0.022, +0.015]	-0.001 [-0.002, +0.000]

None of the pairings improves predictions reliably (0 of 40 checks are reliably better, 1 reliably worse). They are descriptions of how hitters respond, best used at the pitch level in a game plan.

The spin split, pitch by pitch

Table 15: Free Swingers: chase rate (points) above what the batter's and pitcher's own rates on that pitch type predict, by pitcher spin side.

Pitch	Pron. 2020-24	Pron. later	Sup. 2020-24	Sup. later
Changeup/Splitter	+3.2	+6.8	+3.3	+2.7
Curveball	+5.0	+6.8	+5.4	+7.1
Cutter	+4.5	+2.4	+5.1	+5.3
Four-seam	+1.8	+1.0	+1.8	+2.2
Sinker	+5.2	+7.7	+2.9	+1.2
Slider/Sweeper	+2.6	+2.9	+2.6	+3.6

Pitch-level archetype gaps are far more durable than PA-level ones (about 0.75 correlation between periods), but most of that is a batter-type effect: Free Swingers out-chase expectations against *both*

kinds of pitcher. The spin-specific part (how much more a batter type chases supinators than pronators on a pitch type) repeats at $r = 0.45$ for chase and 0.22 for whiff over 30 cells: real, and about a percentage point.

PART 7

Takeaways

What to use

For lineups and bullpens. Use platoon-split career lines, regressed firmly toward league average (or projections). Nothing tested here improves on them.

Head-to-head. Treat it as trivia. Even 15+ prior meetings add nothing measurable on top of career lines.

Archetype tables. They describe real patterns, about a third of which carry forward, but they don't sharpen forecasts. Use them to explain matchups, not to predict them.

Game plans. The most durable matchup physics is at the pitch level: like-on-like batted-ball profiles, matched swing and pitch planes, nibblers vs chasers, chasers vs supinators' shapes. Worth a line in a scouting report, and worth re-testing on 2027 data.

Caveats and next steps

- **Seasons ahead, not games ahead.** Design B freezes everything at the end of 2024; design A at the start of each pair's split season. A game-by-game version would test archetypes properly as an early-season prior for rookies and call-ups.
- **Exploratory matched pairs.** Five pairs, seven outcomes, nine cells each; the highlighted patterns were chosen after the fact. Re-test them on new data before trusting them.
- **Season-level spin.** Savant's Active Spin is one value per pitcher-season; per-pitch spin axes from Hawk-Eye would catch mid-season changes.
- **Approximate noise floor.** With about 17 PAs per pair, the chance-only error in design A is estimated, not known; it is bounded but imprecise.
- **College and summer ball.** Shorter careers and wider talent gaps may behave differently; that needs its own test on the TrackMan data.

Appendix

Table 16: Design B, all methods: skill vs league.

Method	wOBA	xwOBA	K%	BB%	Log loss
League (platoon)	0.00%	0.00%	0.00%	0.00%	0.00%
Career (individuals)	0.13%	0.46%	1.50%	0.51%	0.89%
Career shrunk to velocity / bat-speed tier	0.11%	0.44%	1.52%	0.49%	0.89%
Career shrunk to archetype	0.12%	0.44%	1.51%	0.49%	0.89%
H2H raw	-14.94%	-14.59%	-15.02%	-14.71%	-68.72%
H2H shrunk to league	0.00%	0.02%	0.10%	-0.01%	0.04%
Archetype matchup	0.02%	0.07%	0.40%	0.14%	0.28%
Career + H2H	0.13%	0.46%	1.49%	0.51%	0.89%
Career + archetype	0.14%	0.46%	1.49%	0.51%	0.89%
Career + batter vs pitcher-archetype	0.14%	0.45%	1.49%	0.51%	0.89%
Career + pitcher vs batter-archetype	0.14%	0.46%	1.49%	0.51%	0.89%
Career + batter vs similar pitchers	0.13%	0.46%	1.50%	0.51%	0.89%
All archetype info	0.12%	0.43%	1.51%	0.49%	0.89%
All archetype info + H2H	0.12%	0.43%	1.50%	0.49%	0.88%

Table 17: Design A, all methods: skill vs league.

Method	wOBA	xwOBA	K%	BB%
League (platoon)	0.0%	0.0%	0.0%	0.0%
Career (individuals)	5.4%	13.0%	19.9%	11.0%
Career shrunk to velocity / bat-speed tier	5.6%	12.9%	19.8%	11.0%
Career shrunk to archetype	3.2%	12.4%	19.1%	10.7%
H2H raw	-87.0%	-50.9%	-30.0%	-52.9%
H2H shrunk to league	0.4%	4.4%	10.2%	5.4%
Archetype matchup	0.4%	5.0%	-1.0%	4.8%
Career + H2H	5.4%	12.9%	21.0%	12.2%
Career + archetype	5.6%	13.9%	19.6%	11.0%
Career + batter vs pitcher-archetype	5.6%	14.4%	19.8%	11.0%
Career + pitcher vs batter-archetype	5.6%	13.9%	19.5%	11.0%
Career + batter vs similar pitchers	5.4%	13.0%	20.2%	10.8%
All archetype info	3.2%	12.7%	19.2%	10.6%
All archetype info + H2H	3.0%	12.9%	20.2%	11.5%

Table 18: Regression weights tuned on 2024 from 2020-2023: PAs of prior added (weight on n PAs of evidence = n / (n + k)).

Setting	wOBA	xwOBA	K%	BB%
Career line, batter	600	300	120	240
Career line, pitcher	1,000	450	140	340
H2H on top of career	600	150	150	150
H2H toward league	150	80	40	40
Archetype cell	2,500	5,000	5,000	ignore
Batter vs pitcher-archetype	ignore	1,200	2,500	ignore
Pitcher vs batter-archetype	5,000	ignore	5,000	ignore
Batter vs similar pitchers	600	600	600	2,500
Stacked: cell	ignore	ignore	ignore	ignore
Stacked: batter vs pitcher-archetype	ignore	2,500	2,500	ignore
Stacked: pitcher vs batter-archetype	5,000	ignore	ignore	ignore
Stacked: H2H	600	300	150	150

{**Sources on spin efficiency and pronation/supination.** A. Nathan, `Pitch Movement, Spin Efficiency, and All That'' (The Hardball Times); R. Bunikiewicz,`What You Don't Know About Pronation''; Rapsodo, "Understanding Rapsodo Pitching Data: Spin Efficiency''; Baseball Savant Active Spin leaderboard, 2020–2026.

Code: Code Repo/matchup_archetypes/scripts (run_all.R); outputs in output/. Settings: pair floor 30, spin cut 90%, slots 33/47, pitcher labels pooled, history through 2024, validation season 2024.}